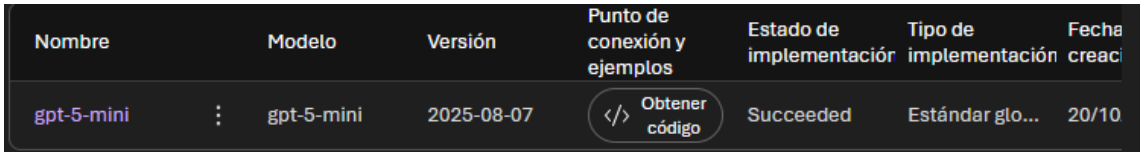
Create a Foundry project and deploy Azure OpenAI models

Creación del Proyecto y selección de los modelos

Seleccionamos gpt-5-mini y lo desplegamos con la configuración predeterminada

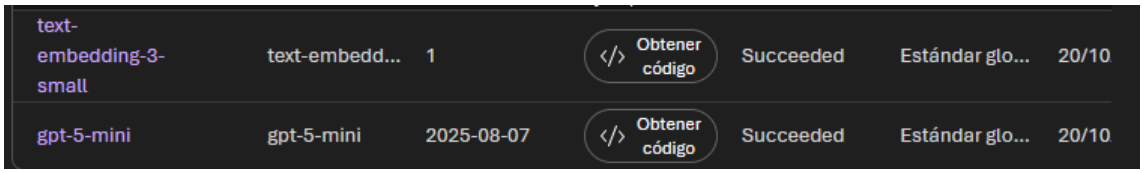


Resultado

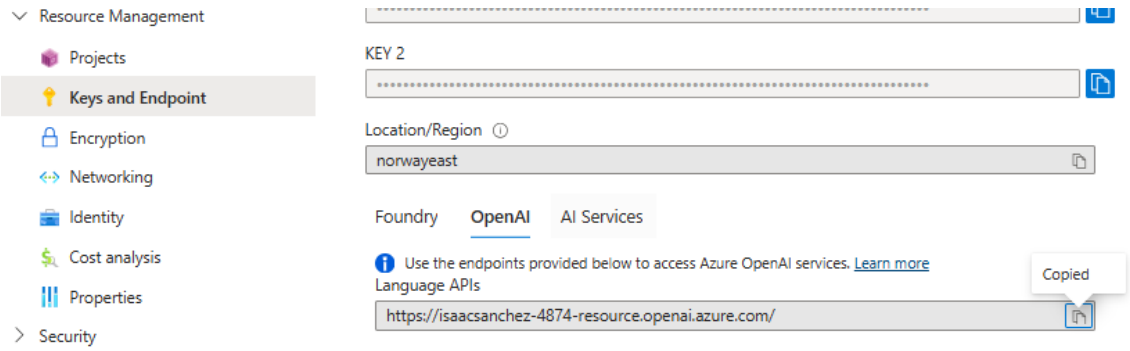


Repetimos con el motor de los embeddings



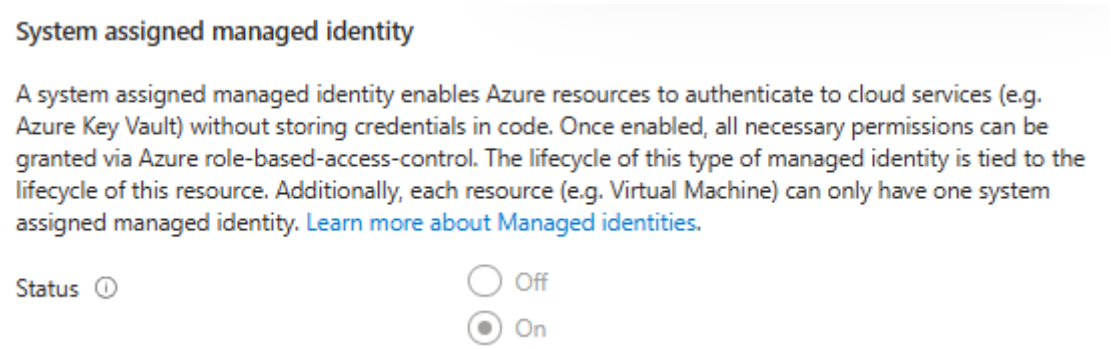


Dentro del recurso entramos en foundry y vamos al apartado de keys and empoint para copiar la url de openAI

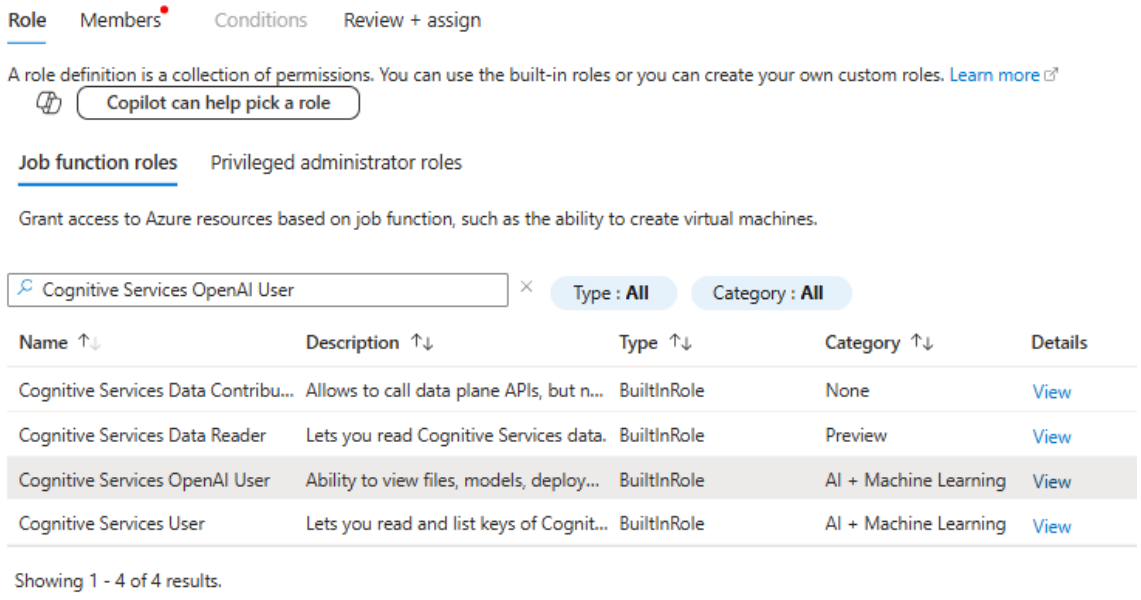


Configure managed identity access

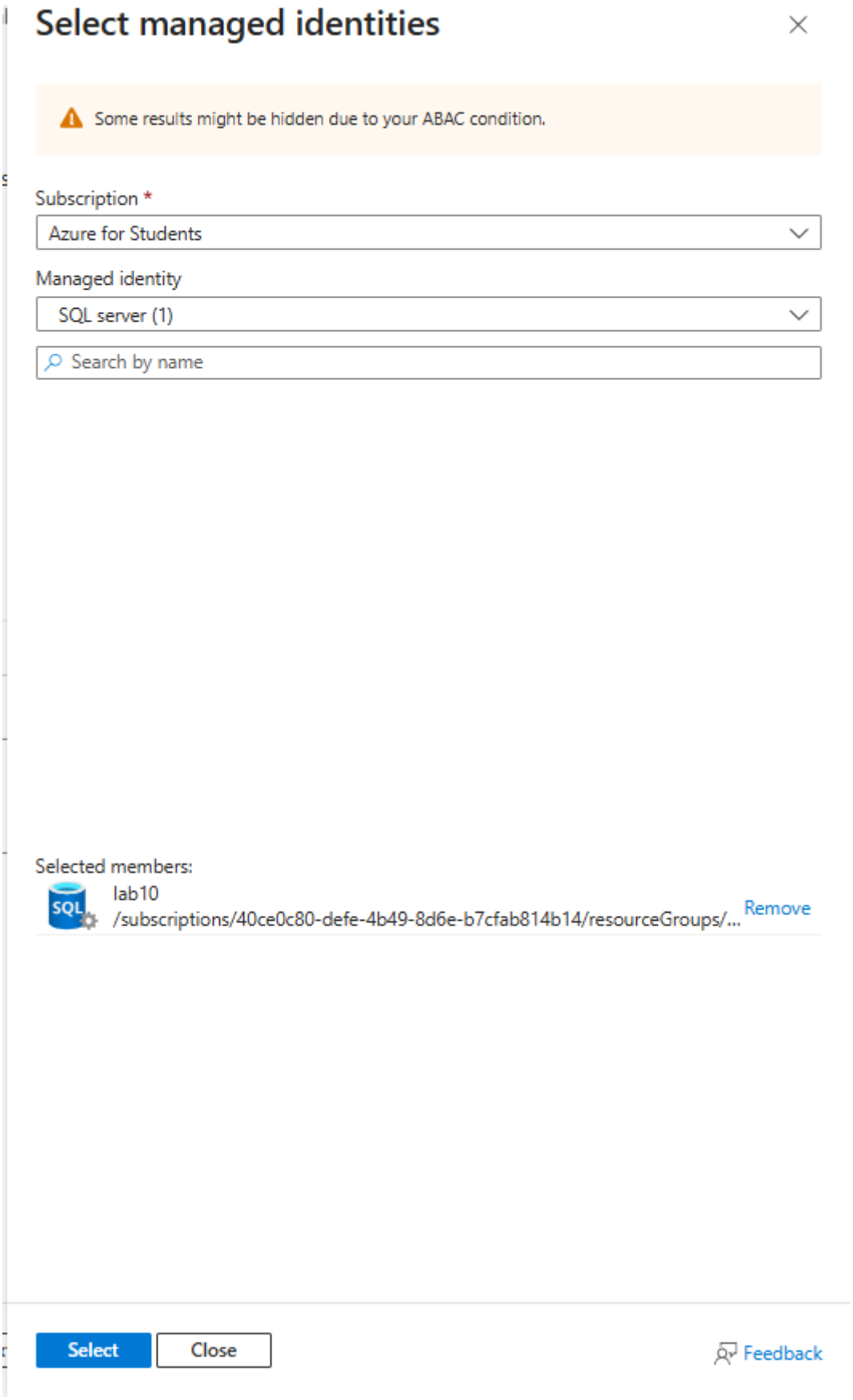
En el portal de azure accedemos al recurso y luego al servidor y acedemos a seguridad identidad y en la sección de maneged identity selecionamos el estado a on



Añadimos el rol de openai

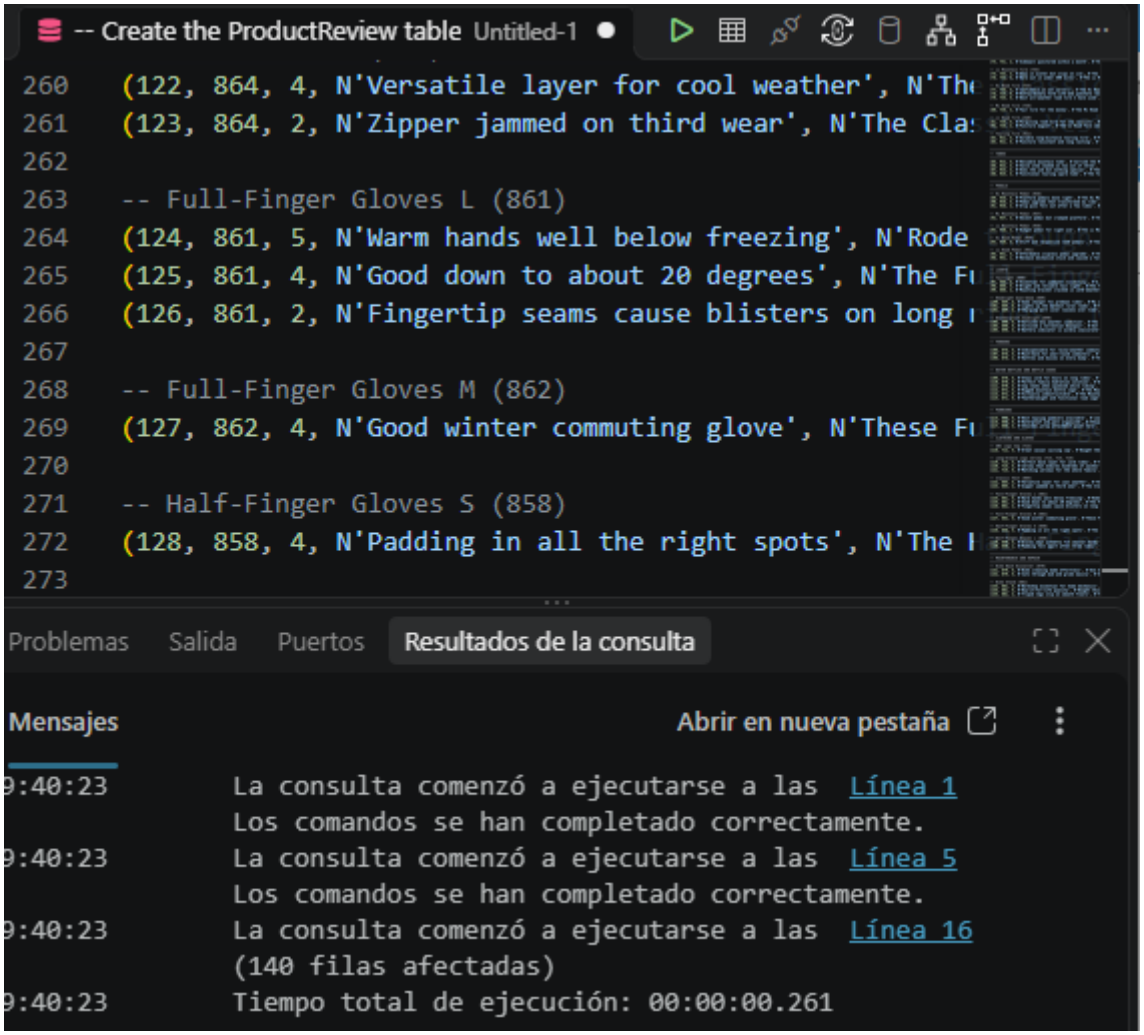


Añadimos como miembro al sqlverver

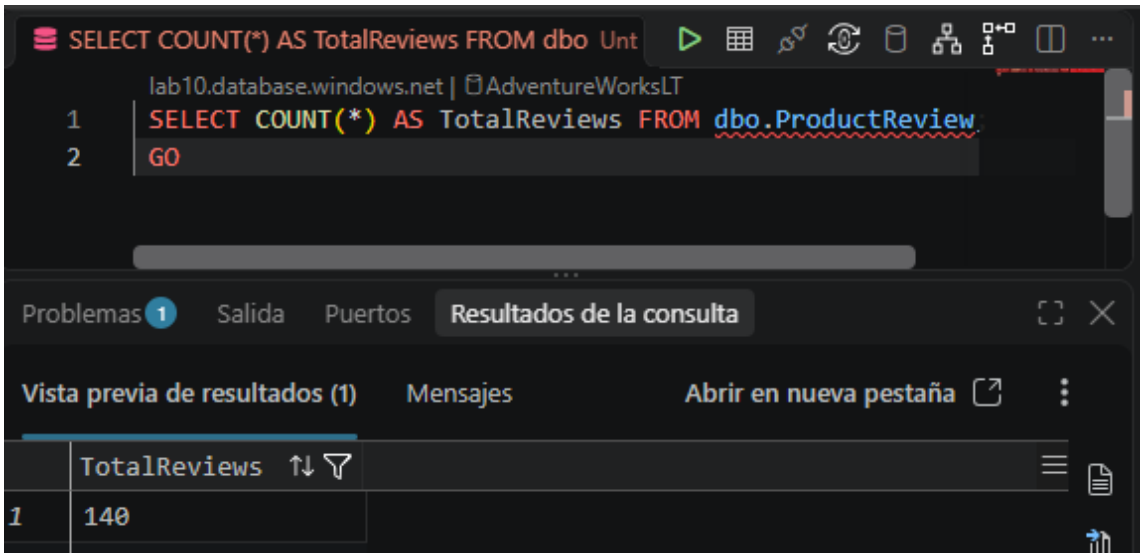


Create the ProductReview table

Nos conectamos a la bbdd con visual estudio y ejecutamos el scrip de creación de tablas



Verificamos si se han creado bien la tablas



Create a database scoped credential and an external model for embeddings

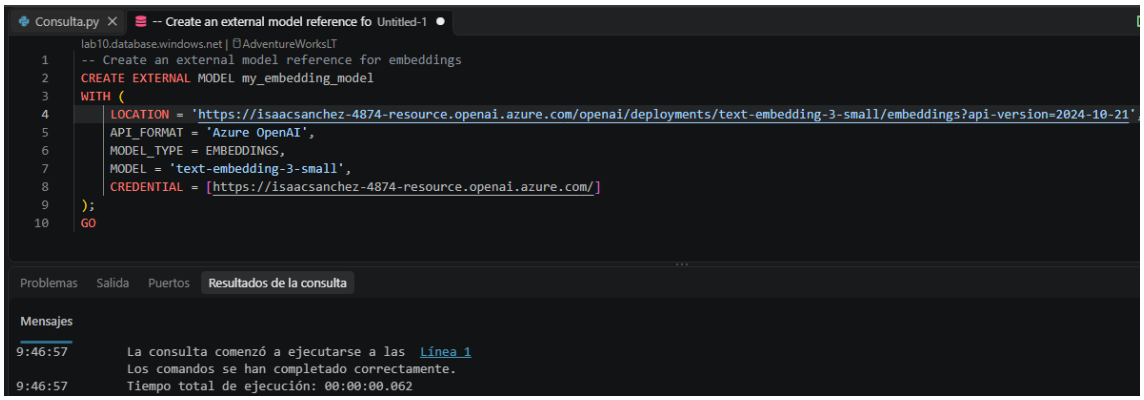
Creamos la credencial para la bbdd

Usando este código `-- Create a database master key if one doesn't exist`

```
 IF NOT EXISTS (SELECT * FROM sys.symmetric_keys WHERE name =
'##MS_DatabaseMasterKey##')
     CREATE MASTER KEY ENCRYPTION BY PASSWORD = '<your strong password
here>';
 GO

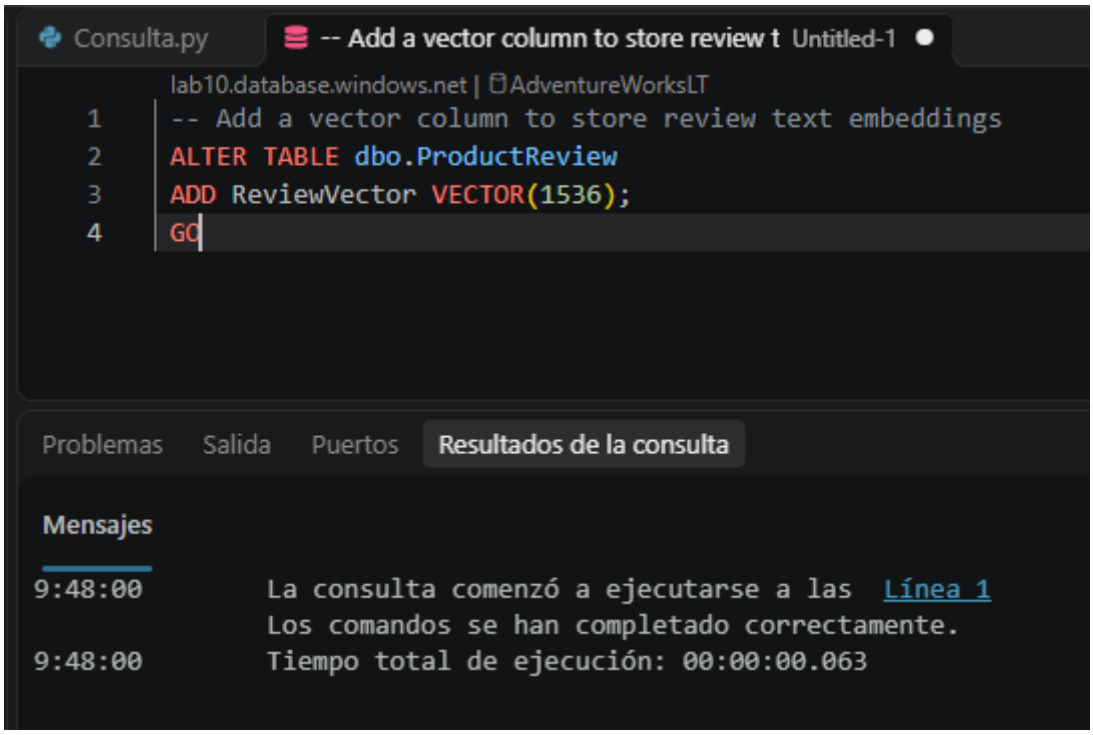
 -- Create a credential using Managed Identity
 CREATE DATABASE SCOPED CREDENTIAL [https://<your-openai-
endpoint>.openai.azure.com]
 WITH IDENTITY = 'Managed Identity', SECRET =
'{"resourceid":"https://cognitiveservices.azure.com"}';
 GO
```

Creamos un modelo externo del modelo de embedding



Add a vector column and generate embeddings

Creamos la columna de los vectores



Generamos los embeddings

Consulta.py -- Generate embeddings in batches to avo Untitled-1

lab10.database.windows.net | AdventureWorksLT

```
-- Generate embeddings in batches to avoid API rate limits
DECLARE @batchSize INT = 30;
DECLARE @rowsUpdated INT = 1;
DECLARE @retryCount INT;
DECLARE @maxRetries INT = 3;

WHILE @rowsUpdated > 0
BEGIN
    SET @retryCount = 0;

    RETRY:
    BEGIN TRY
        UPDATE TOP (@batchSize) r
        SET r.ReviewVector = AI_GENERATE_EMBEDDINGS(
            p.Name + ' - ' + r.ReviewTitle + ': ' + r.ReviewText
            USE MODEL my_embedding_model)
        FROM dbo.ProductReview r
        INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
        WHERE r.ReviewVector IS NULL;

        SET @rowsUpdated = @@ROWCOUNT;

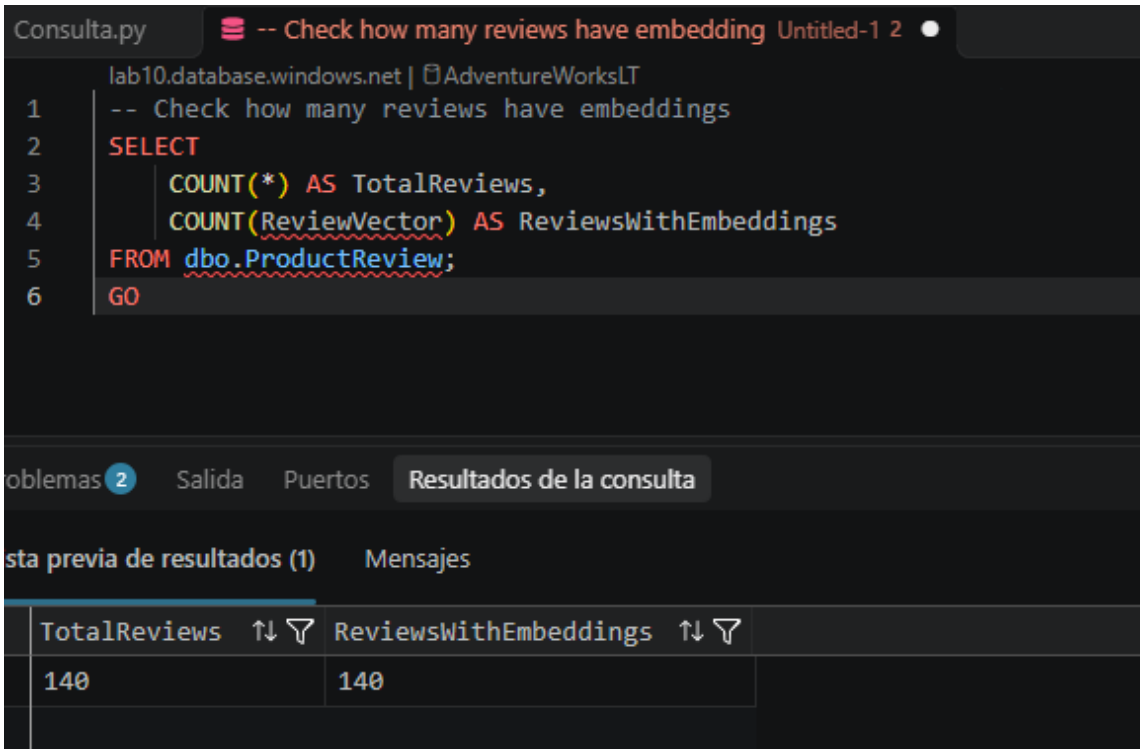
        -- Brief pause between batches to respect API rate limits
        IF @rowsUpdated > 0
```

Problemas Salida Puertos **Resultados de la consulta**

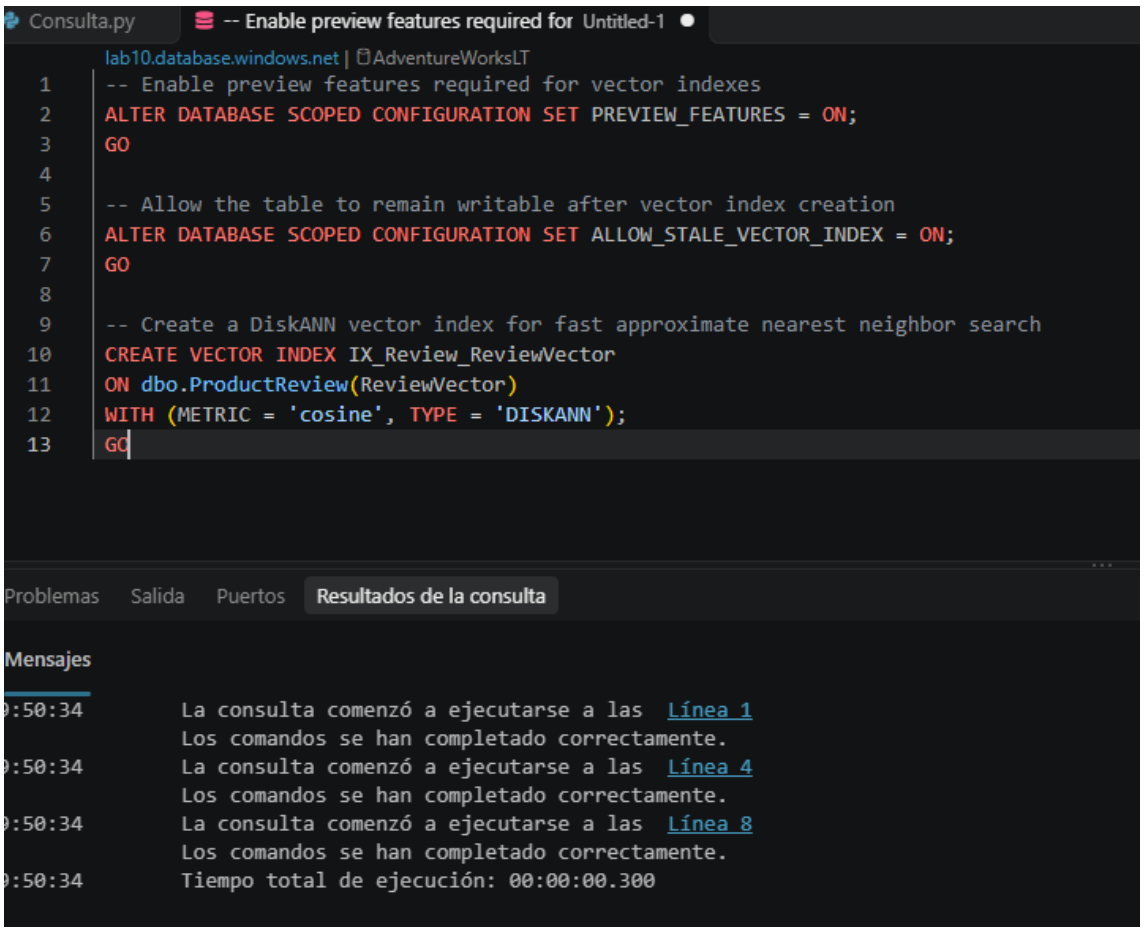
Mensajes

```
9:48:29        La consulta comenzó a ejecutarse a las  Línea 1
               (30 filas afectadas)
               (30 filas afectadas)
               (30 filas afectadas)
               (30 filas afectadas)
               (20 filas afectadas)
               (0 filas afectadas)
9:49:31        Tiempo total de ejecución: 00:01:01.968
```

Verificamos la correcta creacion

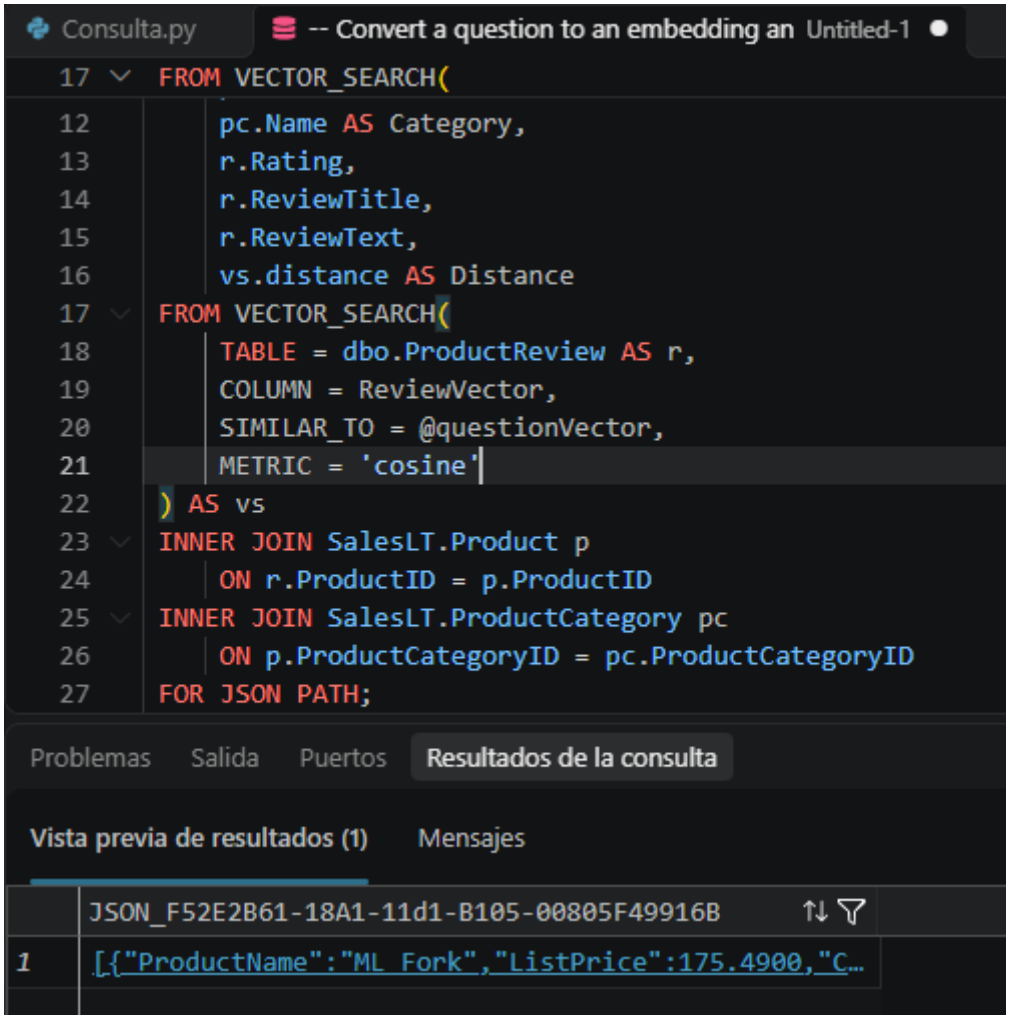


Habilitamos la vista previa y creamos un índice de vectores



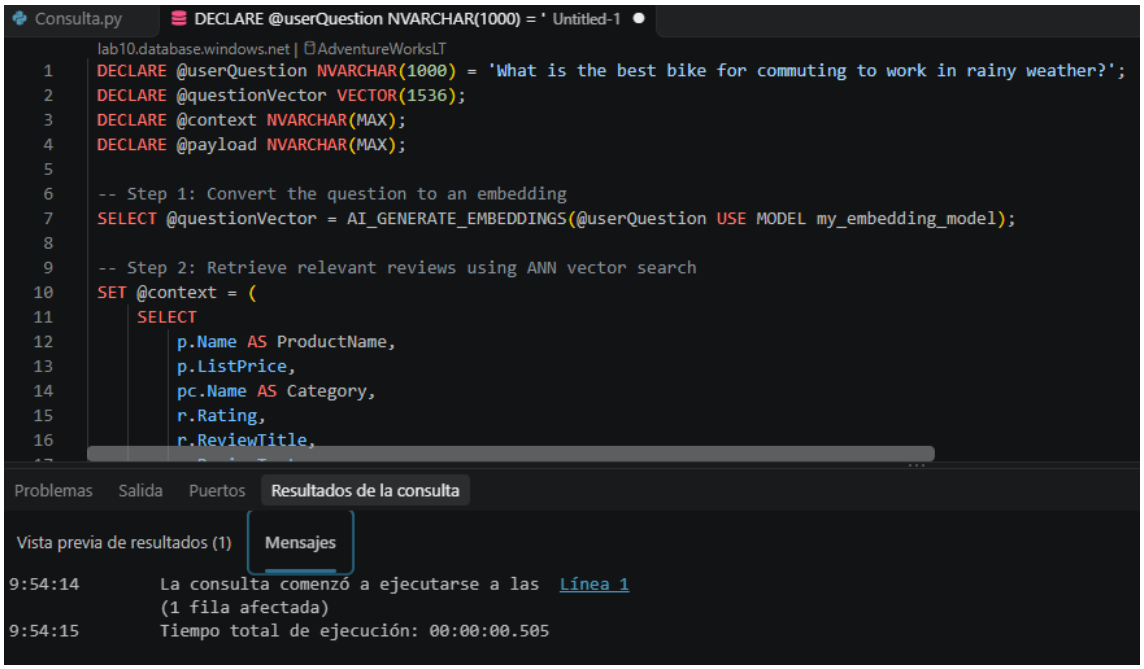
Retrieve data using vector search and format it as JSON context

Usando vector_search realizamos una búsqueda



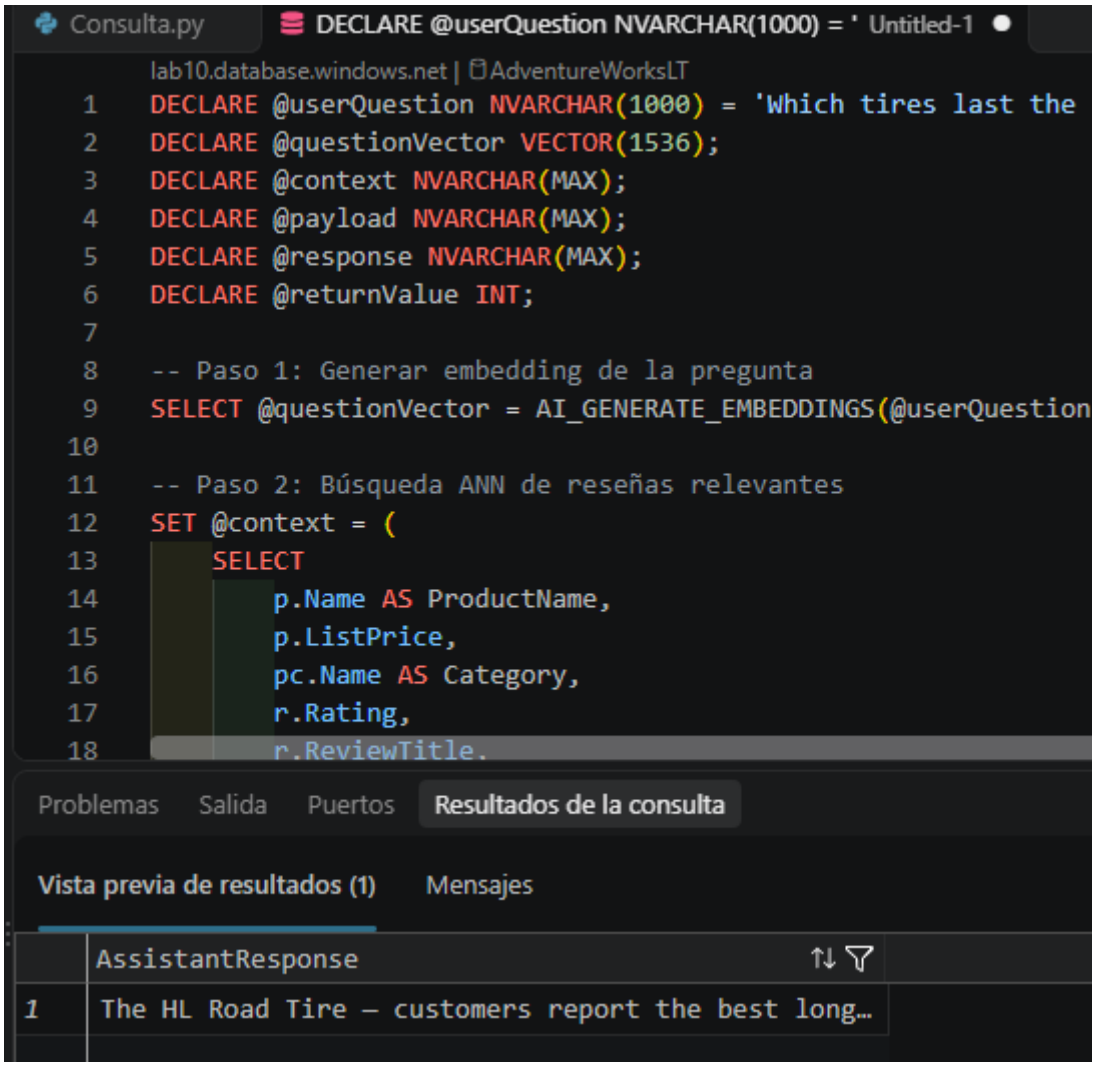
Build an augmented prompt with database context

Ahora, para crear la solicitud aumentada , se combinan los datos recuperados con un mensaje del sistema y una pregunta del usuario . Esta solicitud es la "A" de RAG.



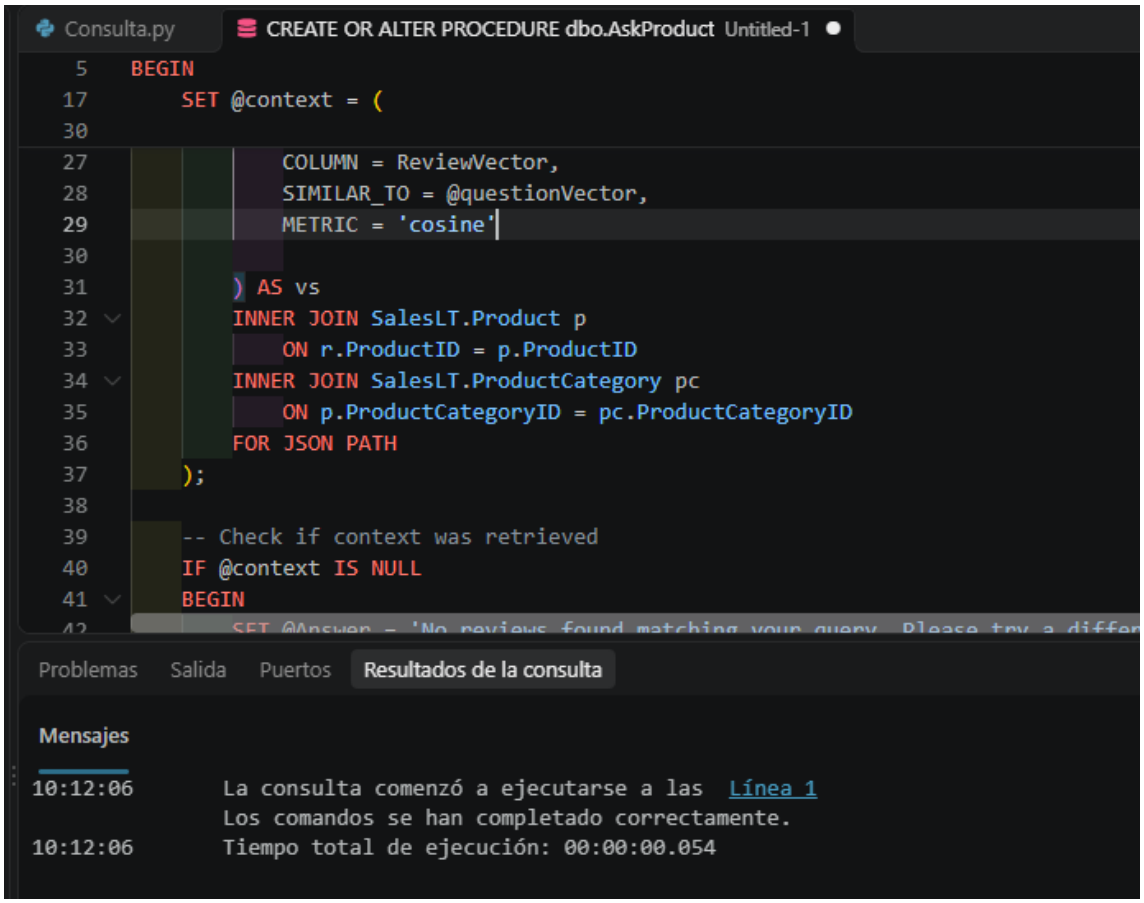
Call the Azure OpenAI endpoint and generate a response

Creamos el endpoint de azure y creamos una respuesta

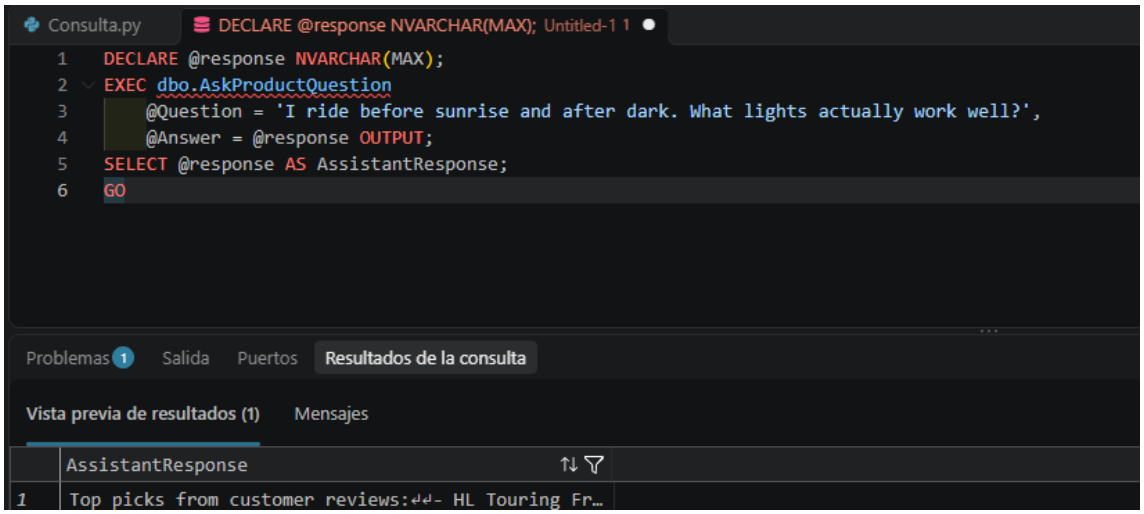


Create a RAG stored procedure

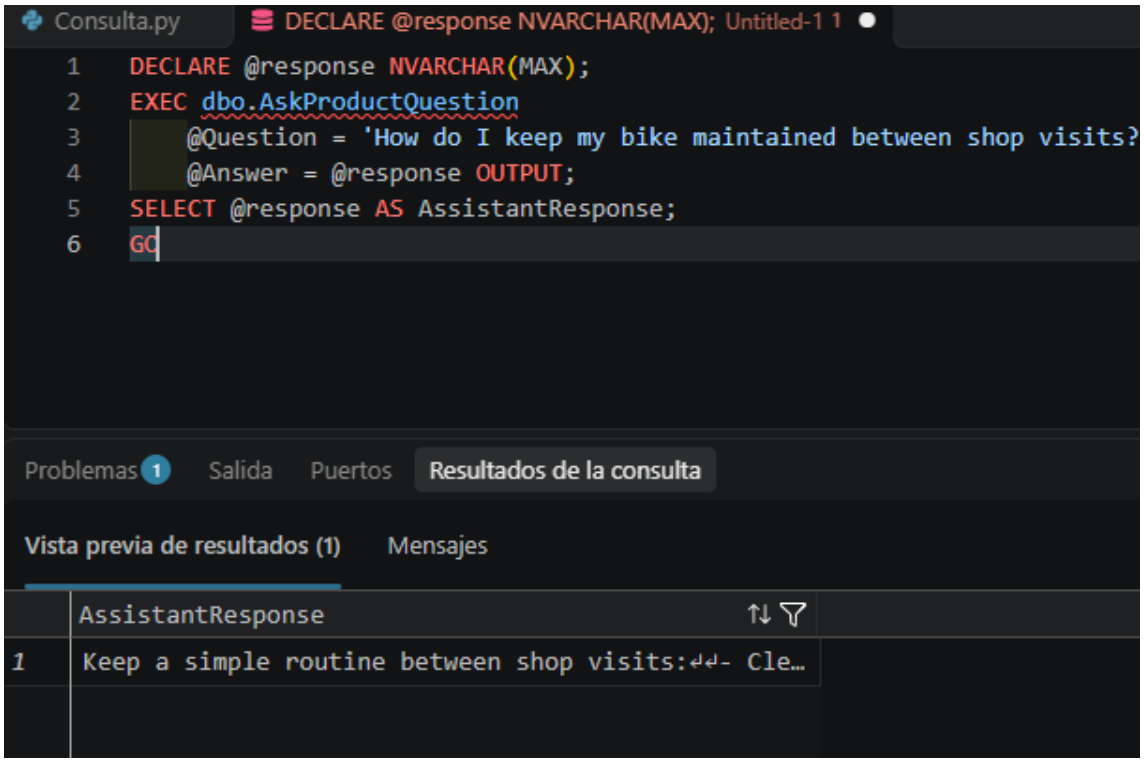
Creamos el procedure



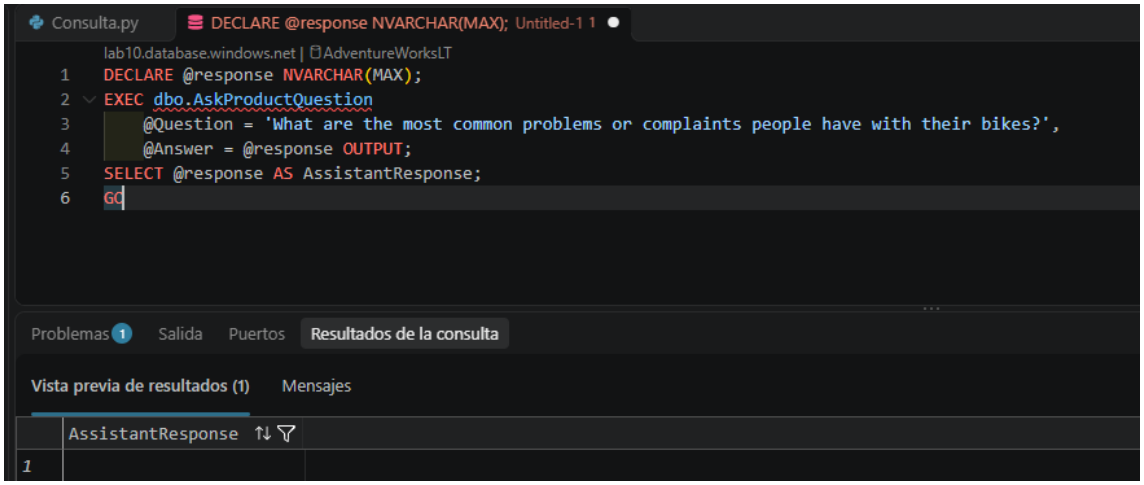
Probar el procedimiento



Prubea 2


Consulta.py
DECLARE @response NVARCHAR(MAX); Untitled-1 1
DECLARE @response NVARCHAR(MAX);
EXEC dbo.AskProductQuestion
@Question = 'How do I keep my bike maintained between shop visits?
@Answer = @response OUTPUT;
SELECT @response AS AssistantResponse;
GO
Problemas 1
Salida
Puertos
Resultados de la consulta
Vista previa de resultados (1)
Mensajes
AssistantResponse
1
Keep a simple routine between shop visits:↵↵- Cle…

Prueba 3


Consulta.py
DECLARE @response NVARCHAR(MAX); Untitled-1 1
lab10.database.windows.net | AdventureWorksLT
DECLARE @response NVARCHAR(MAX);
EXEC dbo.AskProductQuestion
@Question = 'What are the most common problems or complaints people have with their bikes?',
@Answer = @response OUTPUT;
SELECT @response AS AssistantResponse;
GO
Problemas 1
Salida
Puertos
Resultados de la consulta
Vista previa de resultados (1)
Mensajes
AssistantResponse
1